

Q1.

- * A traditional file system stores data in isolated files managed by individual applications, which often leads to high data redundancy, structural dependence, and a lack of built-in security control.
- * A Database Management System (DBMS) acts as a centralized software framework that decouples data from programs, ensuring strictly managed data integrity, concurrent access, and centralized security.
- * A prime scenario requiring a DBMS is a University Course Registration System.
- * If thousands of students try to enroll in popular classes at the exact same second using a simple file system, multiple simultaneous save requests will either overwrite and corrupt the file, or exceed class capacities.
- * A DBMS prevents this by utilizing transactional concurrency control to lock records during a save, ensuring that seat counts remain perfect and zero data is lost.

Q2.

- * A strong entity possesses its own primary key and can exist independently within a database, whereas a weak entity cannot exist without a parent strong entity and lacks a sufficient primary key of its own.
- * The primary key of a weak entity is formed by combining the primary key of its parent strong entity with its own partial key or discriminator.
- * The relationship that binds a weak entity to its strong entity is explicitly called an Identifying Relationship.

Q3.

- * The student phone number must be modeled as a multivalued attribute.
- * Because a single student can possess multiple active numbers simultaneously (such as a personal mobile, home landline, or emergency contact), a standard simple attribute cannot hold these multiple distinct values for a single record without breaking database normalization rules.

Q4.

- * A super key is any combination of one or more attributes that can uniquely identify a row in a table, while a candidate key is a minimal super key, meaning you cannot remove a single attribute from it without losing its uniqueness property.
- * For the relation Student(roll_no, email, name, branch):
 - ① Candidate Keys: {roll_no} and {email} (as both are naturally unique to an individual student).
 - ② Super Keys (that are not candidate keys): {roll_no, name} and {email, branch} (these guarantee uniqueness, but they are not candidate keys because they contain unnecessary extra attributes).

Q5.

- * Attempting to insert a row into the Enrollment table with a student_id that does not exist in the Student table violates a Referential Integrity Constraint (specifically a Foreign Key constraint).
- * This constraint protects against "orphan records," ensuring that the database never contains a broken link pointing to a non-existent entity.
- * The DBMS will automatically reject and abort the insert operation to protect data validity.

Q6.

- * When translating a many-to-many (M:N) relationship to a relational schema, an entirely new table known as a junction or associative table must be created.
- * The primary key of this new table consists of a composite key made up of the foreign keys referencing both participating tables.
- * It cannot simply be represented as a column in one of the participating entity tables because doing so would force a single cell to store a list of multiple values, violating the core rule that all relational database values must be atomic.

Part II - Part A

- * Entities and Key Attributes: Customer (key: customer_id), Restaurant (key: restaurant_id), Order (key: order_id), and MenuItem (key: item_id).
- * Customer to Order Relationship (1:N): A single customer can place multiple orders over time, but any specific order belongs exclusively to one customer.
- * Order to Restaurant Relationship (N:1): Each specific order is linked to exactly one restaurant, though a restaurant will handle many different orders.
- * Order to MenuItem Relationship (M:N): An order can contain multiple different items from the menu, and a popular menu item will show up across many separate orders.
- * Restaurant to MenuItem Relationship (1:N): A single restaurant offers multiple unique menu items, but each specific menu item belongs to that single restaurant's menu profile.

Part II - Part B

- * Multivalued Attribute: Restaurant_Cuisines (or Customer_PhoneNumbers) qualifies because an individual restaurant or customer can have multiple distinct values (e.g., multiple cuisine tags or contact numbers) associated with a single record at the same time.
- * Composite Attribute: Customer_Address qualifies because it is a single concept that naturally breaks down into independent sub-attributes such as street, city, state, and pincode.

Part II - Part C

- * Schema Notation: OrderDetails(order_id, item_id, quantity).
- * Primary Key: (order_id, item_id) combined as a composite primary key.
- * Foreign Keys: order_id referencing Order(order_id) and item_id referencing MenuItem(item_id).
- * Additional Attributes Required: quantity is required to log the specific amount of that menu item requested in that particular order.

Part III Q2

- * The Doctor - Patient relationship is a many-to-many (M:N) relationship because an individual doctor can look after multiple patients at the same time, and a single patient can be treated by multiple different specialists during their stay.
- * This many-to-many relationship creates an associative entity, which can be named Treatment or Consultation. Its composite primary key consists of the combined primary keys of both participating tables: (doctor_id, patient_id).
- * The Patient entity has total participation in both the Ward and Doctor relationships. This is because a person cannot exist as an active patient in a hospital database without being admitted to a physical ward and assigned to a medical doctor for supervision.
- * The Doctor and Ward entities exhibit partial participation. A ward can exist in the system even if it is currently empty, and a doctor can be registered in the hospital database without actively treating a patient.
- * The address of a patient is explicitly modeled as a composite attribute. Breaking it down into smaller sub-components like street, city, and pincode is necessary because it allows the database to easily sort and filter patient records by specific geographic locations.

